

Phishing Analysis Report

AgentTesla Phishing & Data Exfiltration Case

Organization: Globex Manufacturing Ltd.
Analyst Role: SOC Analyst Tier 2
Evidence: 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
Risk Rating: High

1. Executive Summary

On December 4, 2024, Globex Manufacturing Ltd. investigated a suspected phishing email linked to AgentTesla malware activity. The email used a business procurement lure with the subject "PURCHASE QUOTATION" and claimed to request a quotation for attached equipment. Header analysis showed the message was sent from Sertan ÇOKER <sertan@acronas.com.tr>, with an apparent originating IP address of 94.141.120.32. Authentication results showed SPF softfail, DKIM none, and DMARC none, which weakens confidence in the legitimacy of the sender.

Attachment analysis revealed that the email contained a RAR archive. The archive hid a Windows executable named "TECHNICAL SPECIFICATIONS.exe". Static identification showed the payload is a PE32 Mono/.NET Windows executable. The mismatch between the claimed technical specification document and the executable payload indicates malicious intent.

2. Evidence Analyzed

Evidence	Description
Email file	2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
Extracted archive	extracted.rar
Extracted payload	TECHNICAL SPECIFICATIONS.exe
Tools used	grep, emldump.py, unrar, file, md5sum, sha256sum, strings

3. Email Header Analysis

Header Field	Finding
From	Sertan ÇOKER <sertan@acronas.com.tr>
Return-Path	sertan@acronas.com.tr
Reply-To	No Reply-To header observed
Originating IP	94.141.120.32
SPF	softfail
DKIM	none
DMARC	none
Subject	PURCHASE QUOTATION
Date	4 Dec 2024 04:51:17 -0800 / 12:51:17 UTC
Message-ID	<20241204045117.4A43A7B93A5F2488@acronas.com.tr>

The SPF softfail result indicates that the sending IP was not strongly authorized to send mail for the sender domain. DKIM and DMARC were not present, meaning the message could not be strongly authenticated. Although the From and Return-Path values matched, the authentication weakness and malicious attachment make the message suspicious.

```
adiele@adiele-VirtualBox: ~/Downloads/AgentTesla
2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml 2024-12-04-AgentTesla-variant-using-FTP.pcap
adiele@adiele-VirtualBox:~/Downloads/AgentTesla$ grep -i "^From:" 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
From: =?UTF-8?B?U2VydGFuI MOHT0tFug==?= <sertan@acronas.com.tr>
adiele@adiele-VirtualBox:~/Downloads/AgentTesla$
```

Figure 1: Sender, Return-Path and authentication results.

```
adiele@adiele-VirtualBox:~/Downloads/AgentTesla$ grep -iE "(Subject:|Date:|To:|Message-ID:)" 2024-12-04-AgentTesla-variant-malspam-1251-UTC.eml
To: [recipient's email address]
Subject: PURCHASE QUOTATION
Date: 4 Dec 2024 04:51:17 -0800
Message-ID: <20241204045117.4A43A7B93A5F2488@acronas.com.tr>
adiele@adiele-VirtualBox:~/Downloads/AgentTesla$
```

Figure 2: Subject, date, recipient placeholder and Message-ID.

4. Email Content Analysis

The subject “PURCHASE QUOTATION” and body text used a procurement theme. The sender requested a quotation for attached equipment and asked the recipient to review technical specifications in the attached file. This is a business/supplier lure designed to make the recipient open the attachment. Suspicious indicators include weak grammar such as “Dea Sir” and a mismatch between the described document and the actual executable payload.

5. Attachment Analysis

Item	Finding
Outer attachment	RAR archive data, v4, Win32
Inner filename	TECHNICAL SPECIFICATIONS.exe
Inner file type	PE32 executable (GUI) Intel 80386 Mono/.NET assembly for MS Windows
Suspicious strings	txtPassword, txtUsername, btnLogin, frmLogin, pbHidePassword, pbShowPassword, set_PasswordChar

Item	Finding
Likely purpose	Malware delivery and credential theft activity consistent with AgentTesla case context

```

adiele@adiele-VirtualBox: ~/Downloads/AgentTesla
adiele@adiele-VirtualBox: ~/Downloads/AgentTesla$ unrar l extracted.rar
UNRAR 7.00 freeware      Copyright (c) 1993-2024 Alexander Roshal
Archive: extracted.rar
Details: RAR 4
Attributes      Size      Date      Time      Name
-----
..A.....  1896784  2024-12-04  00:02    TECHNICAL SPECIFICATIONS.exe
-----
              1896784                      1
adiele@adiele-VirtualBox: ~/Downloads/AgentTesla$

```

Figure 3: RAR archive listing showing TECHNICAL SPECIFICATIONS.exe.

```

adiele@adiele-VirtualBox: ~/Downloads/AgentTesla
adiele@adiele-VirtualBox: ~/Downloads/AgentTesla$ md5sum extracted.rar
sha256sum extracted.rar
b7635c9cc63619099419c68a2bf0d390  extracted.rar
5c98308c69c84a57214442e2cad9f8f0fcdabab8e6050f9915ac336b6f1d59f0  extracted.rar
adiele@adiele-VirtualBox: ~/Downloads/AgentTesla$

```

Figure 4: File type and hashes for RAR archive and embedded executable.

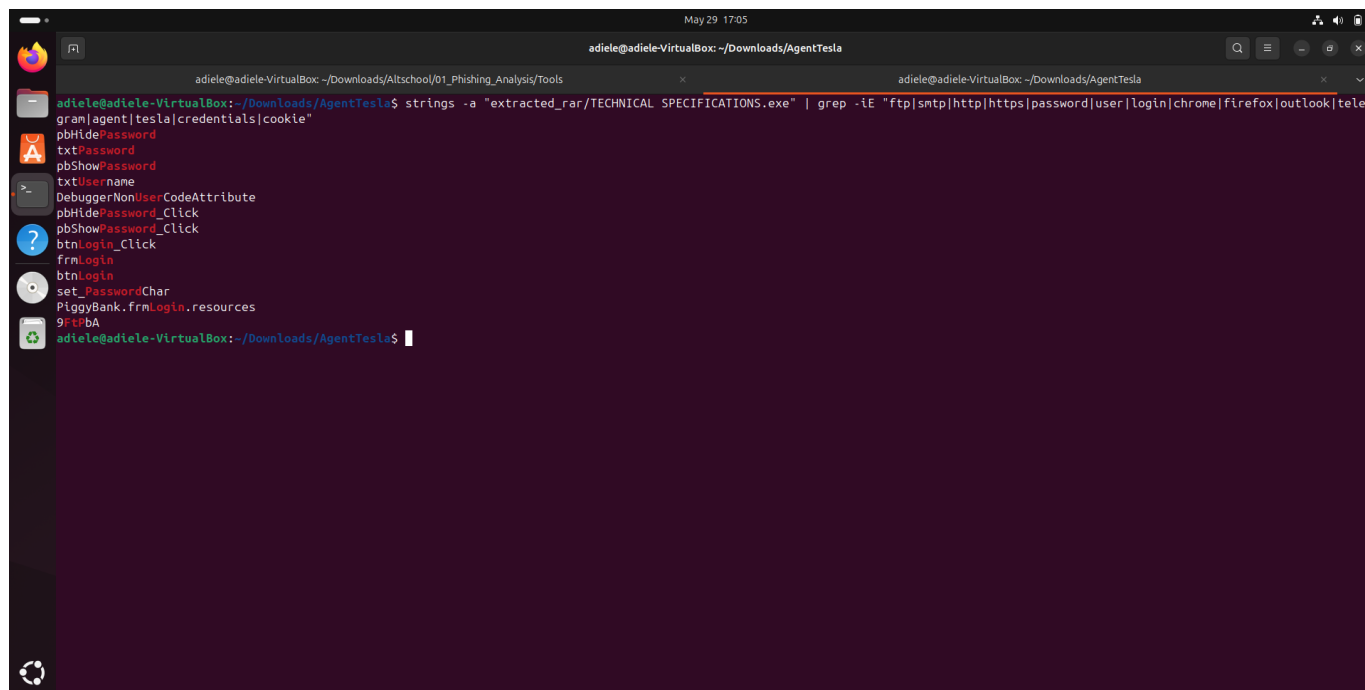


Figure 5: Static strings showing login/password-related indicators.

6. Indicators of Compromise - Phase 1

IOC Type	Value	Source	Notes
Sender Email	sertan@acronas.com.tr	Email header	Appears in From and Return-Path
Display Name	Sertan ÇOKER	Email header	MIME/Base64 encoded sender display name decoded from header
Originating IP	94.141.120.32	Received header	Apparent sending IP visible in the email
Sender Domain	acronas.com.tr	Email header	Domain used in From, Return-Path and Message-ID
Subject	PURCHASE QUOTATION	Email header	Business/procurement lure
Message-ID	<20241204045117.4A43A7B93A5F2488@acronas.com.tr>	Email header	Message identifier
Outer Attachment	extracted.rar	Attachment extraction	RAR archive data, v4, Win32
Payload Filename	TECHNICAL SPECIFICATIONS.exe	RAR listing	Executable disguised as technical specification document
Payload Type	PE32 executable, GUI, Intel 80386, Mono/.NET assembly	file command	Windows executable payload
RAR MD5	b7635c9cc63619099419c68a2bf0d390	md5sum	Hash of extracted RAR attachment
RAR SHA256	5c98308c69c84a57214442e2cad9f8f0fdbab8e6050f9915ac336b6f1d59f0	sha256sum	Hash of extracted RAR attachment
EXE MD5	65feefe926eb3f734b6968b35c23acb3	md5sum	Hash of embedded executable
EXE SHA256	d1b068b826e3a9527cddd09866886caba895f390af930a9b35c027eb1c2db34c	sha256sum	Hash of embedded executable

7. Risk Rating and Initial Hypothesis

Risk Rating: High. The email used a business quotation lure, failed to provide strong sender authentication, and delivered a RAR archive containing a Windows executable disguised as a technical specification document. The initial hypothesis is that the attachment was used to infect the workstation and initiate credential theft, later confirmed by FTP exfiltration activity in the PCAP.

8. Recommendations

Isolate the affected workstation, reset credentials for the impacted user, block the sender domain/IP and payload hashes, search mailboxes for the same sender/subject/attachment, restrict compressed executable attachments, enforce attachment sandboxing, and improve phishing awareness for procurement-themed lures.